

Generation Interconnection System Impact Study Report

For Queue Project Q21-1754

Wheelabrator Portsmouth 138 kV

March 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1754. The Interconnection Customer proposes a 290.6 MW solar plus storage facility interconnecting at the Wheelabrator Portsmouth 138 kV substation in the Virginia area. The request was received on May 21, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 444 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,103,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WHEELABRATOR PORTSMOUTH / VICTORY WAY 138 kV	MICKENS / MIRA VISTA SUBSTATION 345 kV Line	114.2	234	11.9
REEVES AVE / THOLE STREET 138 kV Line	MIRA VISTA / MICKENS 345 kV line	103.0	224	8.2
WHEELABRATOR PORTSMOUTH / REEVES AVE 138 kV Line	MICKENS / MIRA VISTA 345 kV line	102.2	282	8.2
TIPECANOE / SKAGGS ISLAND JUNCTION 2 138 kV	CHANCELLOR ROW / SKAGGS ISLAND JCT 2 345 kV	101.3	295	5.3
TIPPECANOE / OLD CHURCH 138 kV	CHANCELLOR ROW / SKAGGS ISLAND JUNCTION 2 345 kV	100.5	266	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHEELABRATOR PORTSMOUTH / VICTORY WAY 138 kV	Rebuild line with higher-capacity conductor	\$12,752,000	42% (\$5,423,000)
REEVES AVE / THOLE STREET 138 kV	Reconductor line	\$1,787,000	57% (\$1,023,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHEELABRATOR PORTSMOUTH / REEVES AVE 138 kV Line	Reconductor line	\$758,000	57% (\$434,000)
TIPPECANOE / SKAGGS ISLAND JCT 2 SUBSTATION 138 kV	Reconductor line	\$1,016,000	6% (\$61,000)
TIPPECANOE / OLD CHURCH 138 kV	Reconductor line	\$2,721,000	6% (\$162,000)
	Total allocated to project		\$7,103,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.